Q11. Write the characteristic equations for S-R, J-K, D and T FF. Design a flipflop whose characteristic table is shown below, using *J-K* flipflop:

{Characteristics equations ***1 M (0.25 x 4)***, Design of FF: Excitation table ***1 M***, K maps ***1 M*** (***0.5 M for J and 0.5 M for K***)), Logic diagram ***1 M***.} (4)

A	**B**	Q_{n+1}
0	0	1
0	1	Q_n
1	0	0
1	1	$\overline{Q_n}$

Characteristics Equations 1 M (0.25 M x 4):

SR FF*: Q (t+1) = S+R'Q*

JK FF: Q(t+1) = JQ'+K'Q

D FF: *Q(t+1) =D*

T-FF*: Q(t+1) =TQ'+T'Q*

Design of a FF:

Excitation Table (1 M):

A	***B***	Q_n	$Q_{(n+1)}$	***J***	***K***
0	0	0	1	1	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	1	X	0
1	0	0	0	0	X
1	0	1	0	X	1
1	1	0	1	1	X
1	1	1	0	X	1

K map for J (0.5 M):

A \ BQ_n	00	01	11	10
0	1	X	X	0
1	0	X	X	1

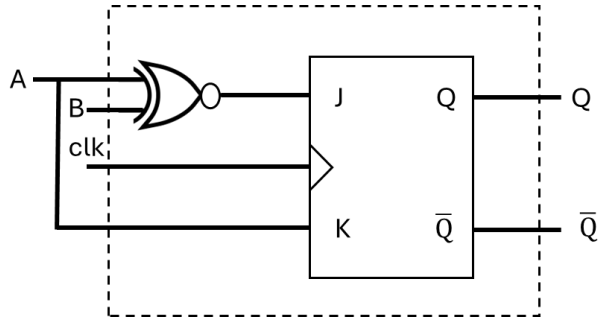
$J = \bar{A}\bar{B} + AB$ = A ⊙ B

K Map for K (0.5 M):

A \ BQ_n	00	01	11	10
0	X	0	0	X
1	X	1	1	X

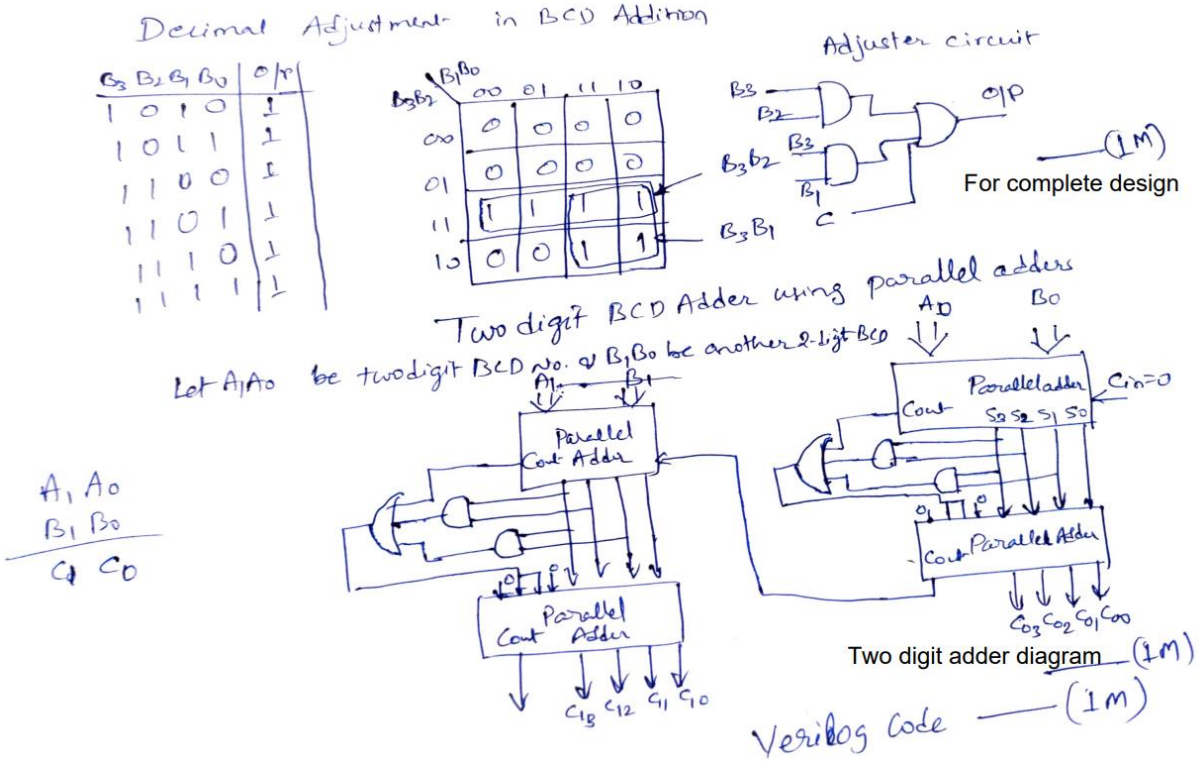
K = A

Logic Diagram (1 M):



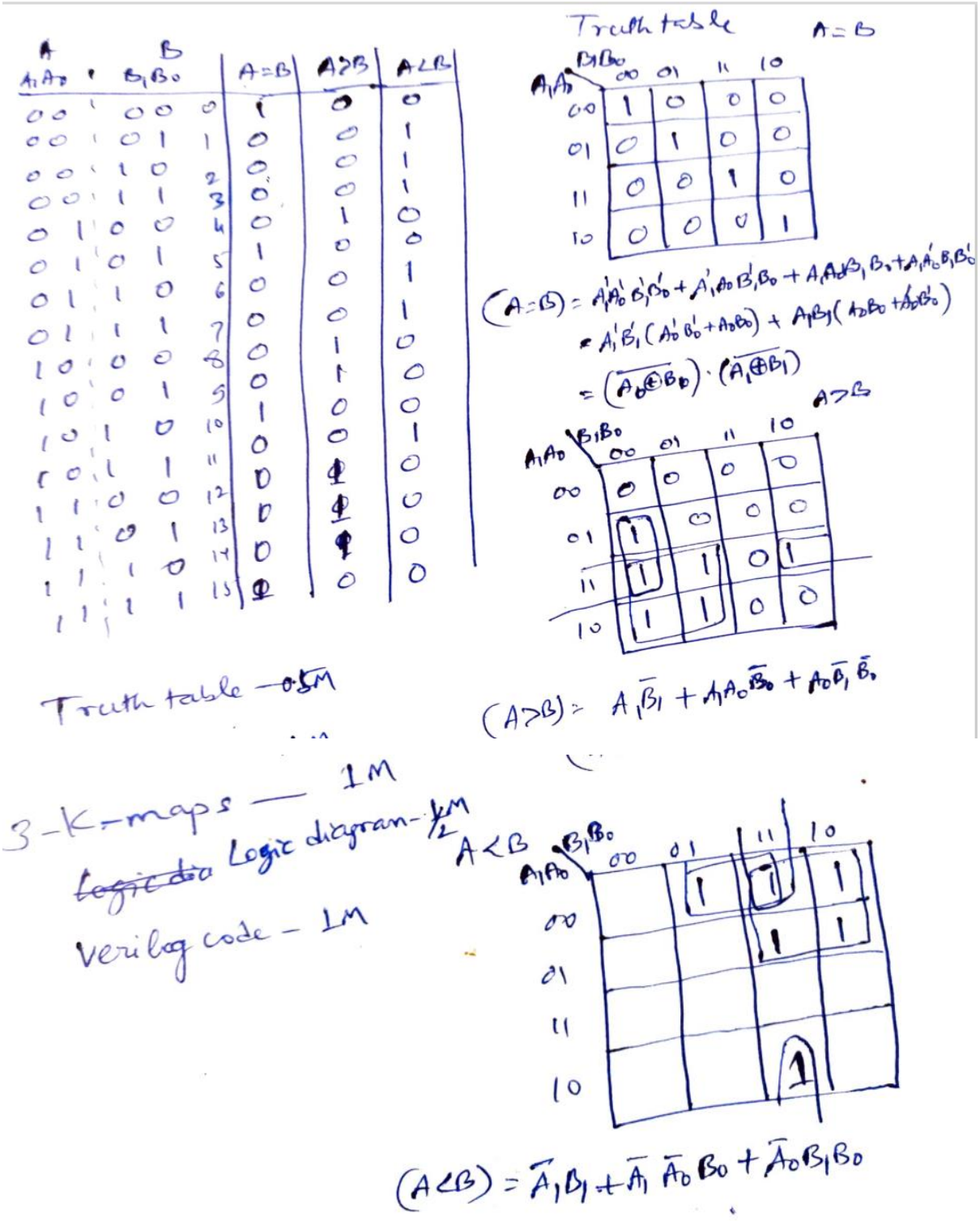
Q12. Design the logic for BCD addition adjustment. Draw the circuit diagram for a two-digit BCD adder using parallel adder. Write a Verilog code for one digit BCD adder using parallel adder structure. (3)

{complete design of BCD addition adjustment: ***1M***, 2 Digit BCD adder using parallel adder: ***1M***, Verilog code: ***1M***}



Q13. Design a two-bit comparator using logic gates and draw the logic diagram. Write a Verilog code for the same. (3)

{Truth table: ***0.5 M***, three K-Maps: **1 M**, Logic diagram: **0.5 M**, Verilog code: ***1 M***}

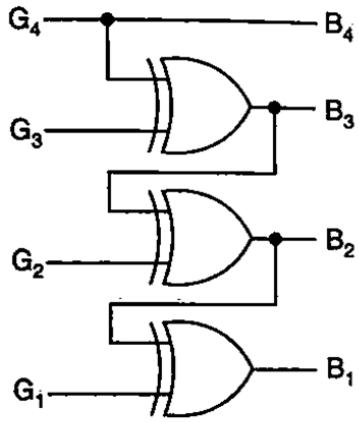


Q14. Design a 4-bit Gray-to-Binary code converter and implement the same using logic gates.

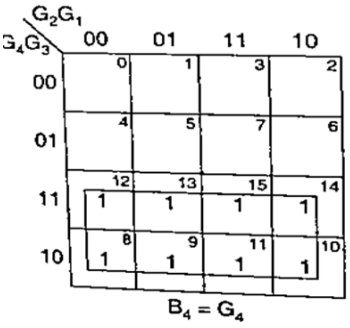
{Conversion Table: ***0.5 M***, Logic Diagram: ***1 M***, K- Map Simplification: ***1.5 M***} (3)

4-bit Gray				4-bit binary			
G_4	G_3	G_2	G_1	B_4	B_3	B_2	B_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

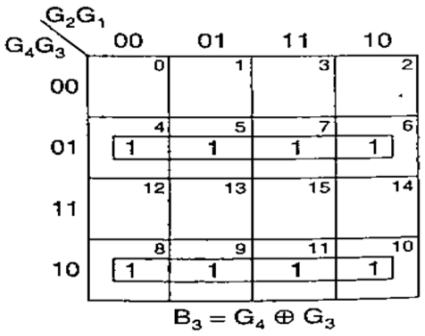
Conversion table



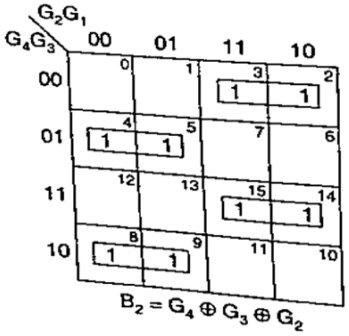
Logic diagram



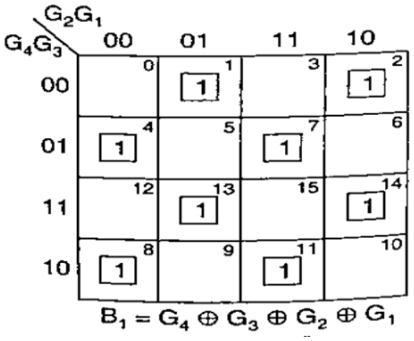
$B_4 = G_4$



$B_3 = G_4 \oplus G_3$



$B_2 = G_4 \oplus G_3 \oplus G_2$



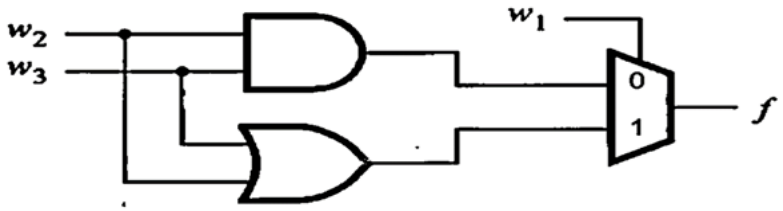
$B_1 = G_4 \oplus G_3 \oplus G_2 \oplus G_1$

Q15. For the function f(w_1, w_2, w_3) = $w_1w_2+w_1w_3+w_2w_3$, use Shannon's expansion to derive an implementation using a 2-to-1 multiplexer and any other necessary gates. { Shannon's expansion: ***2 M*** Logic Diagram: ***1M***} (3)

$$f = w_1w_2 + w_1w_3 + w_2w_3$$

Shannon's expansion using w_1 gives

$$f = \overline{w}_1(w_2w_3) + w_1(w_2 + w_3 + w_2w_3)$$
$$= \overline{w}_1(w_2w_3) + w_1(w_2 + w_3)$$



Q16. A given system has four sensors that can produce an output of 0 or 1. The system operates properly when exactly one of the sensors has its output equal to 1. An alarm must be raised when two or more sensors have the output of 1. Design the simplest logic circuit that can be used to raise the alarm.

{Idea / problem formulation: ***1M***, Drawing the correct K-map: ***1M***, Mapping Ones/zeros and Deriving simplified expression: ***1M***} (3)

Solution: Let the sensors be A, B, C and D. The alarm must be raised when 2 or more of the sensors are high. In K-map we make the state '1' if 2 or more of A, B,C and D are true/High.

AB/CD	CD			
	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	1	1	1
10	0	1	1	1

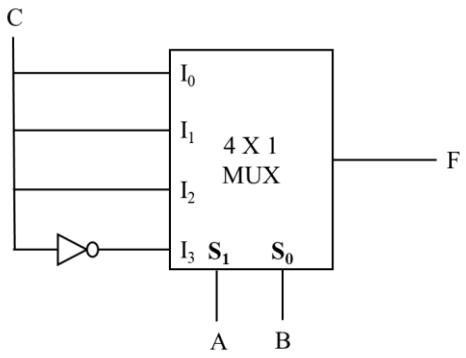
AB

F = AB+ BD + CB + AD + AC+ CD

Q17.Implement F (A, B, C) = Σm (1, 3, 5, 6) using (i) 4X1 multiplexer (ii) 2 x 1 multiplexer and a logic gate (2) {(i) Truth Table: (***0.5 M***), Logic Circuit: (***0.5 M***); (ii) Truth Table: (***0.5 M***), Logic Circuit: (***0.5 M***)}

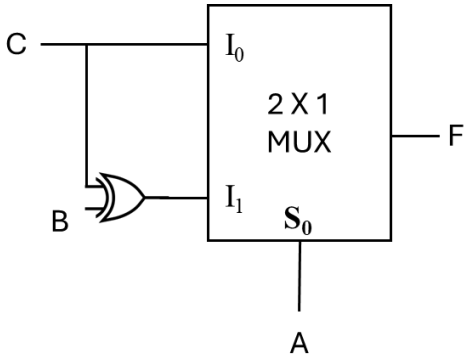
(i) Using 4×1 MUX

S_1	S_0			**Design**
A	**B**	**C**	**F**	
0	0	0	0	F=C » I_0
0	0	1	1	
0	1	0	0	F=C » I_1
0	1	1	1	
1	0	0	0	F=C » I_2
1	0	1	1	
1	1	0	1	F=$\bar{C}$ » I_3
1	1	1	0	

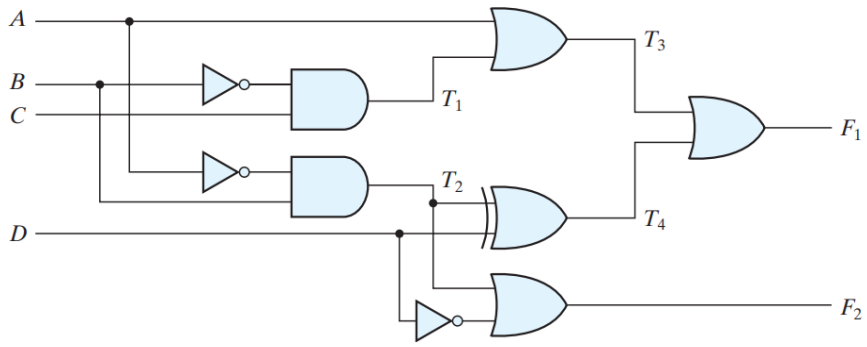


(ii) Using 2×1 MUX

S_0				**Design**
A	**B**	**C**	**F**	
0	0	0	0	F=C » I_0
0	0	1	1	
0	1	0	0	
0	1	1	1	
1	0	0	0	F=B⊕C » I_1
1	0	1	1	
1	1	0	1	
1	1	1	0	



Q18. From the given gate level circuit in Figure .1. derive the Boolean expressions for T1 through T4. Evaluate the outputs F1 and F2 as a function of the four inputs. Find the simplified Boolean expressions for F1 and F2. (2)



T1=B'C **T2=A'B** **T3=A+T1** **= A+B' C** *(0.5 M)*	**T4** = A'B **xor** D = A'BD'+(A+B') D **= A'BD'+AD+DB'** **F2** = T2 + D' **=A'B + D'** *(0.5 M)*	**F1**=T3+T4 = A+B' C + (A'B **xor** D') = A+B' C + A'BD' + AD + DB' = A (1+D) + B' C + A'BD' + DB' = A +B' C + A'BD' +DB' = A + A'BD' + DB' +B' C = **A + BD'+DB' +B' C** *(since A+A'B=A+B)* *(1 M)*

Q19. Derive the simplest Sum-of-products (SOP) and Product of Sum (POS) expressions for the function $f(x_1, x_2, x_3, x_4) = x_1\overline{x_3}\,\overline{x_4} + x_2\overline{x_3}\,x_4 + x_1\overline{x_2}\,\overline{x_3}$ {SOP: ***1M***, POS: ***1M***} (2)

Solution using KMAP.

F= x1. x3'. x4' + x2. x3' .x4 + x1. x2'. x3'

Simplest Sum-of-products (SOP) expression is given by mapping one's

		x3 x4			
		00	01	11	10
x1 x2	00	0	0	0	0
	01	0	1	0	0
	11	1	1	0	0
	10	1	1	0	0

F= x1. x3' + x2. x3' . x4

Hint: Since x+x'=1

Therefore F= x1. (x2+x2'). x3'. x4' + (x1+x1'). x2. x3'.x4 + x1. x2'. x3'. (x4+x4')

Simplest Product-of-Sum (POS) expression is given by mapping zero's with negated variable

